

Scenario 1

1. Scenario: Spelling Correction in a Text Editor

Question: You're creating a simple text editor. How would you correct spelling mistakes in a sentence?

Logical Solution:

- Tokenize the sentence into individual words.
- Use a dictionary or vocabulary to check if each word is spelled correctly.
- For incorrect words, suggest alternatives using edit distance (how many letters need to change).
- Pick the most likely word using word frequency (e.g., "teh" → "the").

2. Scenario: Counting Words in Customer Reviews

Question: You want to analyze customer reviews. How will you count how many times each word is used?

Logical Solution:

- Convert text to lowercase to avoid duplicates (e.g., "Good" and "good").
- Remove punctuation and stopwords (like "is", "the").
- Tokenize the review into words.
- Count the frequency of each remaining word and store in a dictionary format.

3. Scenario: Identifying Questions in Chat Messages

Question: In a chatbot, how do you know if the user typed a question or not?

Logical Solution:

- Check if the message ends with a question mark (?) — a simple rule-based approach.
- Look for wh-words like "what", "why", "how", etc., at the beginning of the sentence.
- Use a classifier trained on examples of questions and statements for more accuracy in complex cases.

4. Scenario: Language Detection for Multilingual Inputs

Question: A user enters text on your website, and you need to detect the language. How would you do that?

Logical Solution:

- Use character-level patterns or word frequency unique to each language.
- Build or use a pre-trained model that has been trained on multiple languages.
- Based on the likelihood of the text matching a known language, classify it (e.g., “Bonjour” is likely French).

5. Scenario: Creating Tags from Blog Articles

Question: You want to create automatic tags for blog articles. How can you generate useful keywords?

Logical Solution:

- Tokenize the text and remove stopwords.
- Identify nouns and noun phrases (e.g., “climate change”, “data analysis”).
- Rank keywords by frequency or importance (e.g., TF-IDF or TextRank).
- Select the top few as tags for the article.

6. Scenario: Convert Text to Lowercase for Uniformity

Question: Why is converting all text to lowercase useful in NLP?

Logical Solution:

- To avoid treating "Apple" and "apple" as different words when they mean the same thing.
- It reduces the number of unique words (helps in vocabulary building and comparison).
- Makes tasks like word counting, matching, and searching more accurate.